

LONDON SCHOOL OF ECONOMICS Department of Economics
EC400 September 2011

Part 2: Mathematics for Microeconomics Examination

Time: 1 hour

Write your name, your class group number, and the name of your class teacher on your script.

This part of your examination counts for 100 marks. Answer all 3 questions.

It is essential that you explain your answers, answers that are correct but not explained do not get high marks.

PTO

1. (20 marks) Consider the following quadratic form:

$$Q(x, y, z) = 5x^2 + 3y^2 + 4yz + 2xz$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(x, y, z) = (x, y, z) A \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Find all the leading submatrices of A .

- (b) (8 marks) Define what it means for the quadratic form $Q(x, y, z)$ to be *indefinite*. How would you be able to assess, in terms of the properties of A , whether $Q(x, y, z)$ is indefinite?
- (c) (8 marks) Is the quadratic form $Q(x, y, z)$ *positive definite*, *negative definite*, *positive semi-definite*, *negative semi-definite* or *indefinite*?

2. (40 marks) Consider the function defined for $x > a$:

$$f(x) = -(\ln(x - a))^2$$

where $\ln(x)$ denotes the natural logarithm of the number x

- (a) (8 marks) Find the *critical points* of $f(x)$ if any exists for $x > a$.
- (b) (8 marks) Are these critical points *local maximizers*, *local minimizers* or *neither*?
- (c) (8 marks) Are these critical points *global maximizers*, *global minimizers* or *neither*?
- (d) (8 marks) Is $f(x)$ *concave*, *convex*, *quasi-concave*, *quasi-convex* or *neither of the above* for $x > a$?
- (e) (8 marks) Let $x^*(a)$ denote the *global maximizer* of $f(x)$ for $x > a$. Use the envelope theorem to show how *maximum value function* responds to a marginal change in a . Also show how the *global maximizer* $x^*(a)$ is affected by marginal changes in a .

3. (40 marks) For $I > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & F(x, y) = x^{1/3}y^{2/3} \\ \text{s.t.} \quad & 3x + y \leq I \\ & x \geq 0 \\ & y \geq 0 \end{aligned} \tag{1}$$

- (a) (8 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither. Which conditions on the concavity/convexity of such functions would suffice to guarantee that the Lagrangian is concave?
- (b) (8 marks) Set up the constrained optimization problem using the Lagrangian function. Can we solve this optimization problem using the Kuhn-Tucker method? Are the Kuhn-Tucker regularity conditions satisfied? Is it possible to solve the same optimization problem with the Lagrange method? Are the Lagrange regularity conditions satisfied?
- (c) (8 marks) Using the Kuhn-Tucker approach, write down the necessary first order conditions for the solution of the constrained optimization problem.
- (d) (4 marks) Are these conditions *necessary* or both *necessary and sufficient*?
- (e) (4 marks) Solve the constrained optimization problem in terms of I .
- (f) (8 marks) Let $x^*(I)$ and $y^*(I)$ be the solution of this problem. Find the slope of such functions with respect to I . Moreover, find how the maximum value function is affected by a change in I .